

# ANDREW VAN OSTRAND

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## EDUCATION

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| <b>Lehigh University</b>                                                                                                                                                                                               | Bethlehem, PA                             |
| <i>Bachelor of Science in Computer Science and Business; GPA: 3.71/4.0</i>                                                                                                                                             | <i>Expected Graduation: December 2026</i> |
| <ul style="list-style-type: none"><li><b>Relevant Coursework:</b> Data Structures &amp; Algorithms, Systems Software, Computer Architecture, Database Systems, Operating System Design, Software Engineering</li></ul> |                                           |

## EXPERIENCE

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| <b>Capstone Software Engineer</b>   <i>Go, Solidity, TypeScript/Node.js, React, Docker, Kubernetes</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Jan 2026 – Present     |
| <i>Oracle</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Bethlehem, PA</i>   |
| <ul style="list-style-type: none"><li>Developing enterprise decentralized identity solution for Oracle Blockchain Platform with policy-based access control, enabling organizations to issue and verify credentials on permissioned Ethereum networks</li><li>Building full-stack verification platform integrating Solidity smart contracts with TypeScript/Node.js APIs and React frontend to enable secure credential issuance and verification workflows</li><li>Engineering modular registry architecture (DID, credentials, revocation, trust) using smart contracts and cryptographic primitives to support secure authentication and authorization workflows</li></ul>                                                     |                        |
| <b>Compiler Research Intern</b>   <i>LLVM, C++, RDMA protocols, Non-Volatile Memory (NVM)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | May 2025 – Present     |
| <i>Lehigh University</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <i>Bethlehem, PA</i>   |
| <ul style="list-style-type: none"><li>Developed Serenity compiler using LLVM-based optimizations for NVM and RDMA operations, achieving 7× throughput improvement over SSD architectures and 60% reduction in storage infrastructure costs</li><li>Designed developer-friendly API abstractions that reduce RDMA implementation complexity from weeks to hours, enabling broader adoption of high-performance computing capabilities</li></ul>                                                                                                                                                                                                                                                                                     |                        |
| <b>Software Engineer Intern</b>   <i>Node.js, React, SQL Server, Socket.io, REST APIs</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Dec 2025 – Jan 2026    |
| <i>Vomar Products</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <i>Los Angeles, CA</i> |
| <ul style="list-style-type: none"><li>Architected full-stack production tracking platform serving 50+ employees, eliminating manual status checks and reducing job lookup time by 75% through real-time WebSocket updates across 166+ concurrent jobs</li><li>Engineered automated ERP integration system syncing JobBOSS data every 5 minutes via RESTful API, eliminating manual data entry and providing single source of truth for job status, priority, and progress tracking</li><li>Deployed role-based authentication system with JWT tokens, replacing error-prone network drive workflow and eliminating document version conflicts that previously caused 15+ hours/week of rework across manufacturing teams</li></ul> |                        |

## PROJECTS

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| <b>Prof Pair</b>   <i>AWS Comprehend, SageMaker, S3, Flask, Python</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <ul style="list-style-type: none"><li><b>Won Best Use of AI Award</b> at university hackathon for building intelligent professor-student matching platform analyzing 10,000+ course reviews to connect students with compatible professors across two universities</li><li>Built end-to-end ML pipeline on AWS SageMaker leveraging Comprehend NLP for sentiment analysis and topic extraction, processing multi-dimensional review data to generate personalized faculty recommendations</li><li>Architected scalable Flask API with S3-backed storage designed to support university-scale deployment for 8,500+ students</li></ul> |
| <b>Truth Trail</b>   <i>Solidity, SKALE, GCP, Gemini AI, Vertex AI, React</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <ul style="list-style-type: none"><li>Engineered blockchain-based evidence management system providing immutable audit trails for forensic investigations, solving chain-of-custody vulnerabilities in traditional evidence tracking</li><li>Designed dual-layer architecture integrating SKALE blockchain for tamper-proof storage with GCP-hosted AI classification system using Gemini/Vertex AI for automated evidence categorization</li><li>Reduced evidence verification time by 60% through smart contract automation while maintaining cryptographic proof of authenticity for legal compliance</li></ul>                    |

## TECHNICAL SKILLS

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| <b>Languages:</b> JavaScript/TypeScript, Python, Java, C/C++, SQL, Solidity                                                              |
| <b>Frontend:</b> React, HTML/CSS, Socket.io, REST APIs                                                                                   |
| <b>Backend:</b> Node.js, Express, Flask, JWT Authentication, WebSocket                                                                   |
| <b>Cloud &amp; Data:</b> AWS (S3, EC2, Lambda, SageMaker, Comprehend), GCP (Vertex AI, Compute Engine), SQL Server, Oracle DB            |
| <b>Developer Tools:</b> Git, Docker, Linux/Unix, Postman, Maven, JUnit, JDBC, LLVM                                                       |
| <b>Specializations:</b> Full-Stack Development, Machine Learning/AI, System Design, Compiler Optimization, Blockchain, Real-Time Systems |